

1. Let $M_{3,3}$ be the vector space of 3×3 matrices, and let A and D be the matrices

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 1 \\ 2 & 0 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 2 & -2 & 2 \end{bmatrix}$$

Let V be the subset of $M_{3,3}$ given as

$$V = \{C \in M_{3,3} \mid CA = 0\}$$

where 0 is the matrix of all 0's. Is B an element of V ? Argue for your answer. Now assume we are given some matrix D in $M_{3,3}$ and form the subset

$$W = \{C \in M_{3,3} \mid CD = 0\}$$

Is W a subspace of $M_{3,3}$? Argue for your answer. *Hint:* Note that unlike the case of V , I haven't told you what D is. For example, D could be A , and in that case $V = W$, but D could also be any other matrix. If you are unsure of how to approach this problem, try to first check if V is a subspace, then use what you have learned for the more general case of W . In your written solution you should then only include your solution for W .

2. For this problem, consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 1 \\ 3 & -1 \end{bmatrix} \qquad C = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

Recall that these can be considered as vectors of the vector space $M_{2,2}$ of 2×2 matrices. Are the three matrices A, B, C linearly independent when considered as vectors in $M_{2,2}$? Write out what precisely what it means for them to be linearly independent and argue for your answer.

3. For the matrix A below, compute the following: The rank, the dimension of the row space, the dimensions of the column space, a *basis* for the nullspace (not just the dimension of the nullspace):

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix}$$

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a) to check if matrix B is in subset V , we have to check if Matrix B satisfy the condition $C \in M_{3,3} \mid C \cdot A = 0$
 $B \in M_{3,3}$ so if $B \cdot A = 0$ then B is in

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 2 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 1 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1-3+2 & 2-2 & -1-1+2 \\ 0 & 0 & 0 \\ 2-6+4 & 4-4 & -2-2+4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$B \cdot A \longrightarrow 0 \text{ matrix}$

condition is satisfied, so matrix B is part of the subset V

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Is V subspace of Vector Space?

1.) is zero vector in V ?

$$C \in M_{3,3} \mid C \cdot A = 0$$

$$\text{is } C \text{ is zero vector, then } C \cdot A = 0 \Rightarrow \begin{matrix} 0=0 \\ 0=0 \end{matrix} \Rightarrow \vec{0} \in V \quad 1/3 \checkmark$$

2.) Closed under addition

if for every pair of vectors in V , sum is also in V

so for $\vec{w}, \vec{v}$:

$$(\vec{w} + \vec{v}) \cdot A = 0$$

$$\vec{w} \cdot A + \vec{v} \cdot A = 0 \Rightarrow \vec{w} + \vec{v} \in V$$

$$0 + 0 = 0 \Rightarrow 2/2$$

3.) Closed under the multiplication

$k \in \mathbb{R}$ (scalar)

$$(k \cdot C) \cdot A = 0$$

$$k(C \cdot A) = 0$$

$$k \cdot 0 = 0 \Rightarrow k \cdot C \in V \quad 3/3$$

So V is subspace

W will be subspace if it will satisfy these conditions

1.) $\vec{0}$ in set

2.) Closed under addition

3.) Closed under multiplication

$$W = \{C \in M_{3,3} \mid CD = 0\}$$

if $D = \vec{0}$, then W would be entire space $M_{3,3}$

if $D = A$, then $W = V$ the same subspace

else we will have to check 3 conditions to find out

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Recall that these can be considered as vectors of the vector space $M_{2,2}$ of 2×2 matrices. Are the three matrices A, B, C linearly independent when considered as vectors in $M_{2,2}$? Write out what precisely what it means for them to be linearly independent and argue for your answer.

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Linearly independent vectors (matrices) can not be written in terms of other

$$a \cdot A + b \cdot B + c \cdot C = 0$$

$$a \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 3 & -1 \end{bmatrix} + c \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = 0$$

$$\begin{cases} a & 2c = 0 \\ 2a + b - c = 0 \\ 3b - c = 0 \\ -a - b + c = 0 \end{cases} \rightarrow \boxed{b = \frac{c}{3}}$$

$$b - a - \frac{c}{3} + c = 0$$

$$a = \frac{-c + 3c}{3}$$

$$\boxed{a = \frac{2}{3}c}$$

$$\frac{2c}{3} + 2c = 0$$

$$\boxed{c = 0}$$

$$\boxed{a = 0}$$

$$\boxed{b = 0}$$

only solution is if $a, b, c = 0 \Rightarrow$ Linearly INDEPENDENT

3. For the matrix A below, compute the following: The rank, the dimension of the row space, the dimensions of the column space, a basis for the nullspace (not just the dimension of the nullspace):

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix}$$

1.) Get $\text{REF}(A)$

$$\begin{aligned} A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix} &\xrightarrow{\substack{R_2 \rightarrow R_2 + (-4) \cdot R_1 \\ R_3 \rightarrow R_3 + (-3) \cdot R_1}} \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 0 & -2 & -8 & -4 & 1 \\ 0 & -2 & -8 & -5 & 2 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 \cdot (-1) \\ R_3 \rightarrow R_3 + (-1) \cdot R_2}} \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 0 & 2 & 8 & 4 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + 4 \cdot R_3} \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 0 & 2 & 8 & 4 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \\ &\xrightarrow{\substack{R_2 \rightarrow R_2 \cdot \frac{1}{2} \\ R_3 \rightarrow R_3 \cdot (-1)}} \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 0 & 1 & 4 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + (-2) \cdot R_2} \begin{bmatrix} 1 & 0 & -5 & 3 & \frac{1}{2} \\ 0 & 1 & 4 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + (-3) \cdot R_3} \begin{bmatrix} 1 & 0 & -5 & 3 & \frac{1}{2} \\ 0 & 1 & 4 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \end{aligned}$$

$$\text{REF}(A) = \begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & 1 & 4 & 0 & \frac{3}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Row Space

(NOT NULL ROWS)

REF(A)

$$\begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & 1 & 4 & 0 & \frac{3}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Non zero rref(A) rows = row space RS(A)

$$\vec{v} = [1, 0, -5, 0, 2]$$

$$\vec{w} = [0, 1, 4, 0, \frac{3}{2}]$$

$$\vec{u} = [0, 0, 0, 1, -1]$$

number of vectors = 3 = row rank

Vectors $\vec{v}, \vec{w}, \vec{u}$ form base for row space of A

Now I can say that Rank and Column Space is 3 because RANK = # of vectors that forms base for RS =
= # of vectors that forms base for CS

Column Space

(Columns of A where Pivots in rref(A) are)

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix} \quad \text{REF}(A) = \begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & 1 & 4 & 0 & \frac{3}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Pivot columns of rref(A) forms vectors in A. that vectors are base for Column Space

$$\vec{v} = \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$$

number of vectors = 3 = Column rank

$$\vec{w} = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} 3 \\ 8 \\ 4 \end{bmatrix}$$

Vectors $\vec{v}, \vec{w}$ and $\vec{u}$ form base for Column Space of A

$$\text{column rank} = 3$$

$$\text{row rank} = 3$$

$$\text{RANK}(A) = 3$$

Null Space of A

$$N(A) = A \cdot \vec{x} = 0$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix} \cdot \vec{x} = \vec{0}$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 4 & 6 & 4 & 8 & 9 \\ 3 & 4 & 1 & 4 & 8 \end{bmatrix} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \xrightarrow{\text{ref}(A) \mid 0} \begin{array}{c} \text{RREF}(A) \\ \begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & 1 & 4 & 0 & \frac{3}{2} \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \end{array}$$

$$\left. \begin{array}{l} x_1 - 5x_3 + 2x_5 = 0 \\ x_2 + 4x_3 + \frac{3}{2}x_5 = 0 \\ x_4 - x_5 = 0 \end{array} \right\}$$

$$\boxed{x_4 = x_5 = s} \quad s \in \mathbb{R}$$

$$x_2 + 4x_3 + \frac{3}{2}s = 0 \quad ; \quad \boxed{x_3 = t, t \in \mathbb{R}}$$

$$\hookrightarrow x_2 + 4t + \frac{3}{2}s = 0$$

$$\boxed{x_2 = -\frac{3}{2}s - 4t}$$

$$x_1 - 5t + 2s = 0$$

$$\boxed{x_1 = -2s + 5t}$$

$$\vec{x} = \begin{bmatrix} -2s + 5t \\ -\frac{3}{2}s - 4t \\ t \\ s \\ s \end{bmatrix} = \begin{bmatrix} -2 \\ -\frac{3}{2} \\ 0 \\ 1 \\ 1 \end{bmatrix} s + \begin{bmatrix} 5 \\ -4 \\ 1 \\ 0 \\ 0 \end{bmatrix} t$$

$$s, t \in \mathbb{R}$$

Null space of A